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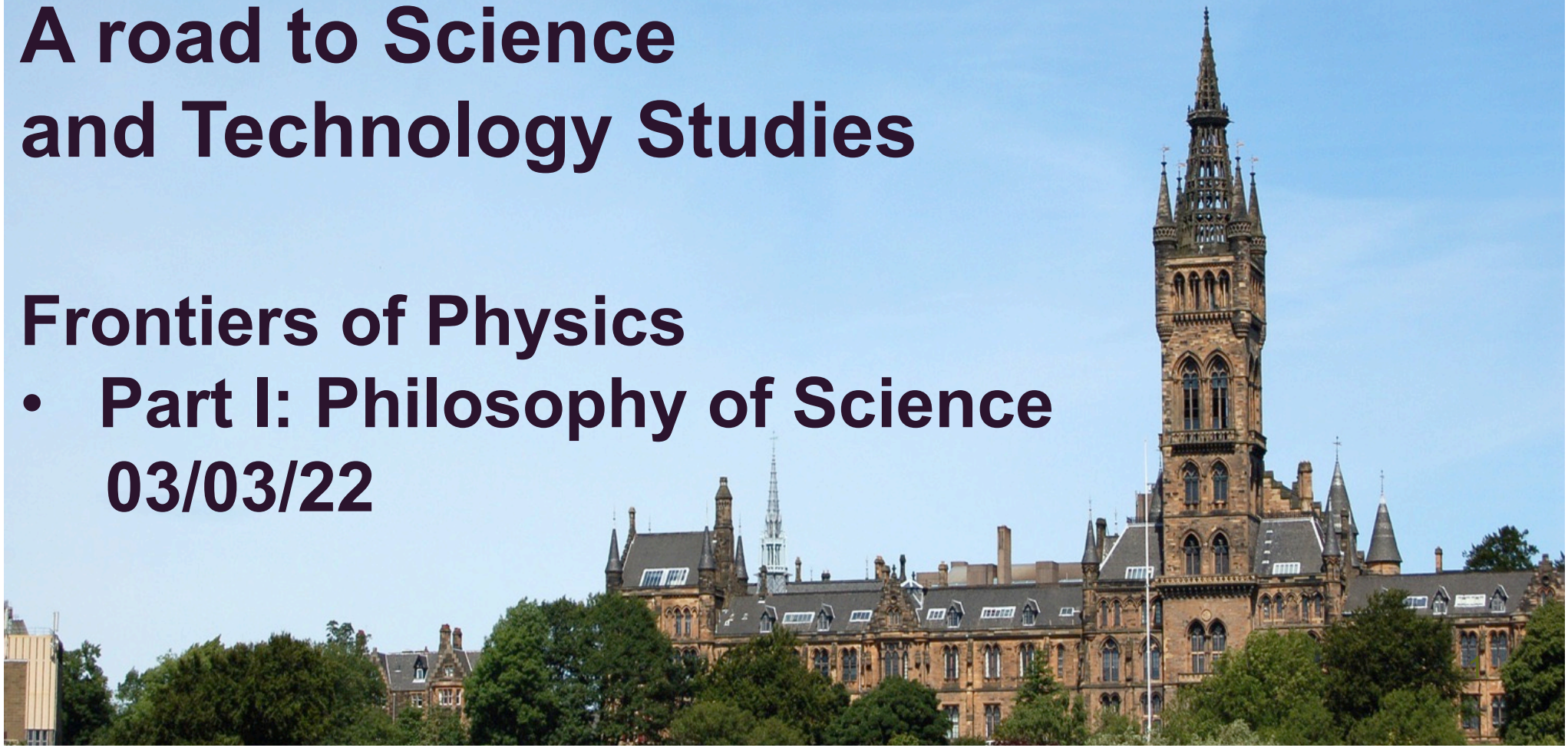
Kenneth Wraight

Let's Talk About Science:

A road to Science and Technology Studies

Frontiers of Physics

- Part I: Philosophy of Science
  03/03/22



Where are we going today?

Motivation
- Importance of science in society
- Sorting science (demarcation problem)
- Exceptionalism: USP

Analytical Approach
- Some common sense
- Importance of empiricism
- Search for Method

Inductive model
- (Deduction)
- Induction
- Riddles of Induction

Hypothetico-Deductive model
- Hypothetico-Deduction
- Confirmation
- Falsification

Science WTF?

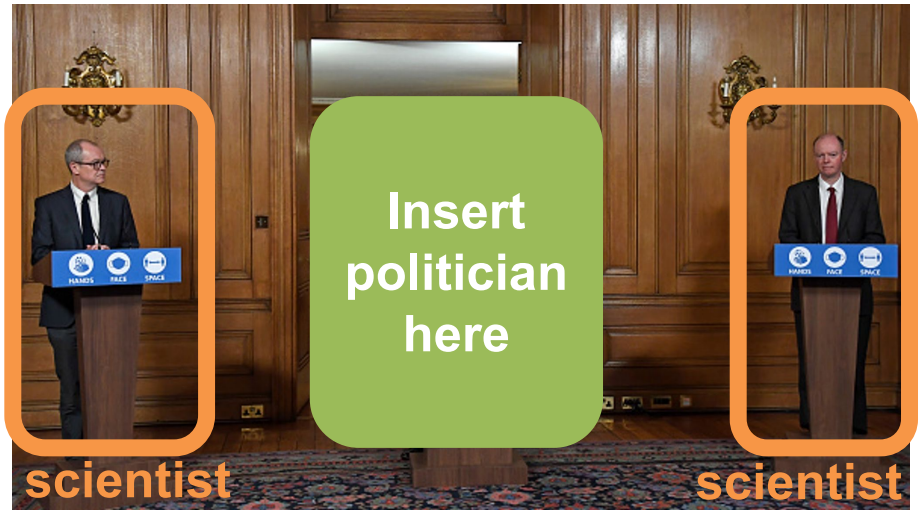
We can't start at the beginning or call a timeout on *Science*

- Science has been around a while: history is important
- Science is ongoing: current state is important (probably not final)

Does the who/what/why of *Science* matter?

If you think it's a question worth asking: Welcome to STS!
If you aren't so sure: here are some issues

- Politics: position of authority in society
- Economics: investment from governments and private companies
- Society: trust (or not) from citizens



Most other human endeavors do not have such high esteem

- So you might ask: **What makes Science special/different**?

Deciding difference

The issue of separating science form non-science is known as the *Demarcation Problem* (spoiler: still a problem)

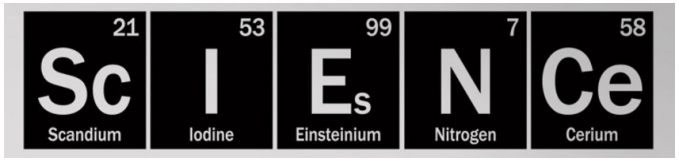


The Demarcation Problem: the separation of science from other activities, a.k.a. **policing** science

- E.g. Astronomy in and Astrology out

Answers to *Demarcation* depend on your opinion of what Science does

- **Aims**:
  truth, predictive accuracy, explanation, utility
- **Methods**:
  induction, confirmation, falsification, paradigmatic eras, anarchy
- **Products**:
  facts/knowledge, techniques, technology



Can we agree on **some basic elements** to Science?

- Characteristics shared by (most) things called science

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Characteristics

Take your intuitively favourite set of scientific disciplines…

Are there *Necessary and Sufficient* conditions for science

- ***Necessary Condition***: essential requirement
  e.g. Breaking eggs is a necessary condition to make omelets

Science⇒X

Is there a property (set) of science that must be present?

- ***Sufficient Condition***: enough on its own
  e.g. Smoke is a sufficient condition to conclude fire

X⇒Science

Is there unique property (set) of science?

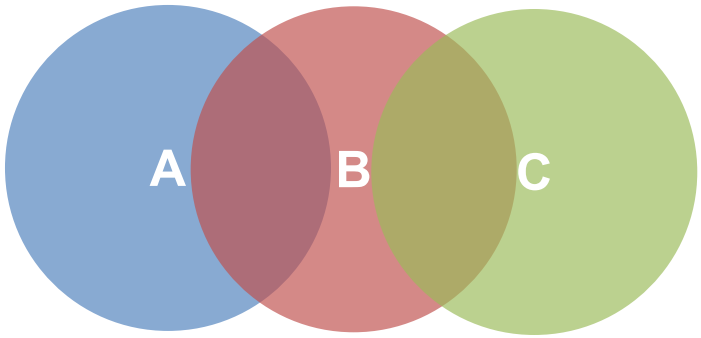
- ***Equivalence Condition***: A *Necessary and Sufficient*
  e.g. A bachelor is an unmarried male

X⇔Science

Is there a uniquely definitive property (set)?

- ***Family Resemblance***:

overlapping groups without universal property (set)
  e.g. “game” is not uniquely defined



What are the overlapping of sets?

Take a moment…

Take a couple of minutes to consider basic properties of science

- Is there a property (set) you consider crucial: necessary/sufficient
- Is there a definitive property (set) you can define - equivalence
- Is there nothing that covers all the intuitive cases

What is Science like?



What is Science <u>not</u> like?



Deciding Reality – an analogy

In sport…
Referees have ***ontological authority*** – decide matters based on ***epistemic privilege*** – special characteristics



Epistemic privilege: best position to decide
- Well placed on the field
- Impartial
- Trained in interpretation



Ontological Authority: decide what is real
- Decide legitimate events: foul/fair
- Decide quality of events: score/no score
- Decide quantity of events: points



Status from Authority: Responsibility & Recognition
- ➢ Power – inside institution (game)
- ➢ Respect – outside of institution (game)

NB Authority is undermined when alternative privilege exists (instant replay!)

Common Sense Science

Science has large amount of *ontological authority* in society

- What is the *epistemic privilege* of Science?

What is Science like?

Empirical

- Led by experience

Knowledge generating

- Testible - predictive
- Objective - unbiased
- Certain – strong inferences
- Cumulative – built over centuries
- Stable – reliable knowledge
- Rational – structure of inference

What is Science <u>not</u> like?

Non-Empirical

- Innate/inherited knowledge

Not knowledge generating

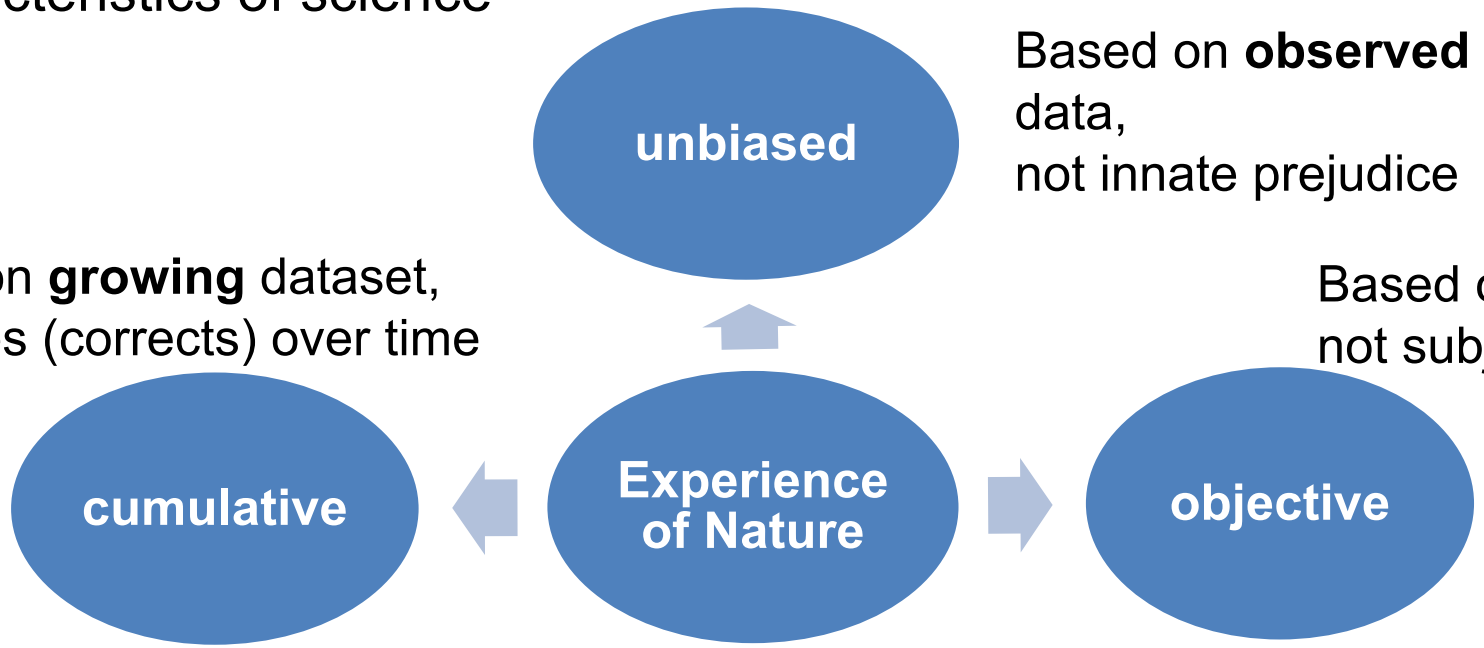
- Approximate/vague
- Subjective
- Uncertain
- Stagnant/decreasing
- Chaotic
- Irrational

Key part in the ***common sense*** image of science is ***empirical*** characteristic

- **Focus on experience of Nature is important…**

Empirical Importance

In the common sense picture, the *empirical* element underlies other characteristics of science



Empirical element brings much of the *epistemic privilege*

Experience to Knowledge

Steps to *Scientific* knowledge:



Empiricism: a theory of knowledge (epistemology) based on experience

Science records/collects experience

- This is not particular to science - lots of activities use experience as a guide

Some process occurs to **convert experience into knowledge**

- Theories are more than the list of events that have taken place

A *common view* is that science has a **method** for this

- **Is this method the special ingredient**?

Past analyses of science see potential in this avenue for science USP

Let's start here!

A Method of Science

What is the **recipe/algorithm/logic** of scientific knowledge

- Progressing from particular experience to general knowledge

More questions:

- Is the method sufficient or necessary?
  - Can it be used for *demarcation*?
- Is it universal across all sciences?
  - Are different methods appropriate for different contexts (*pluralism*)?
- Can it be replicated?
  - Can old/other subjects be made scientific?

These questions raise issues involving

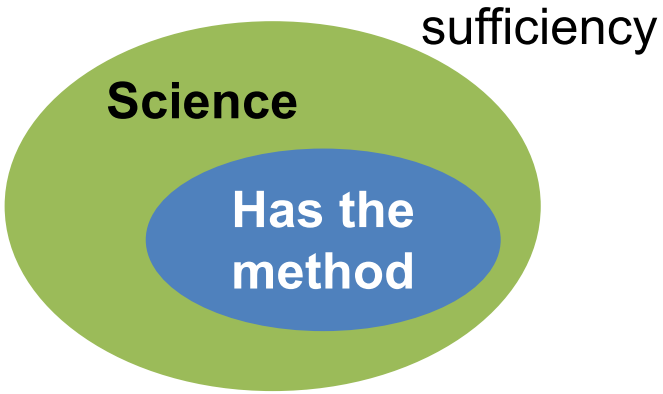
- *characteristics* of science – not just listing sciences
- *quality* of science - grading knowledge

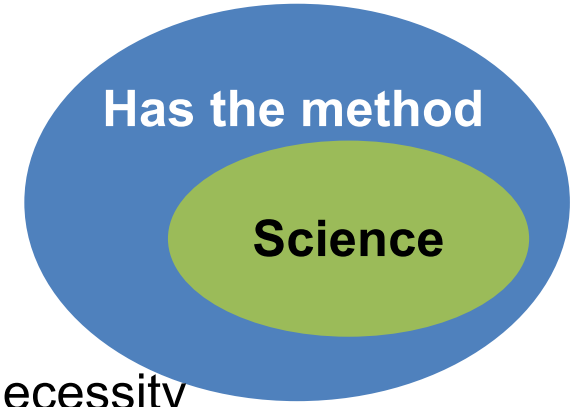
} **Analytic** task, Not empirical

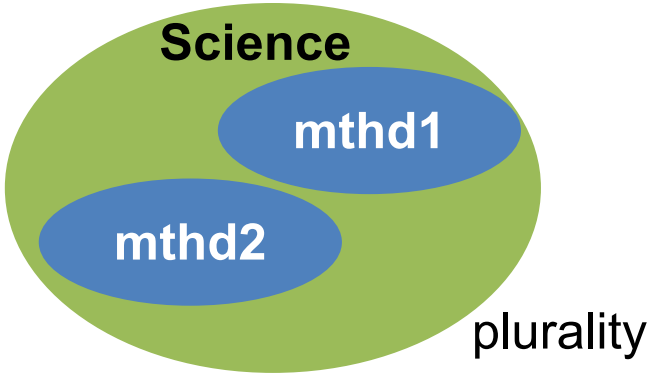
This is the task taken up by Philosophy of Science

Take a moment…

Do any of the diagrams match your intuition for method in science?







More questions:

- Is the method sufficient or necessary?
- Is it universal across all sciences?
- Can it be replicated?

A Brief History of PoS

Philosophy and *Philosophy of Science* go way back

- Term scientist only coined ~1833 by William Whewell
- But many of the old heroes of modern science (especially physics) thought of themselves as *Natural Philosophers*

Philosophy about *Nature* (*Natural Philosophy*) goes back as far as *Philosophy*

- Separation of subjects is relatively modern in the history of ideas

Eventually the empirical part of the *New Science* (ca. 18th century) became more emphasised and popular until (empirical) *Science* diverged

- Seen again with Economics and Psychology later on
- After people continued as philosophers *or* scientists (though exceptions)

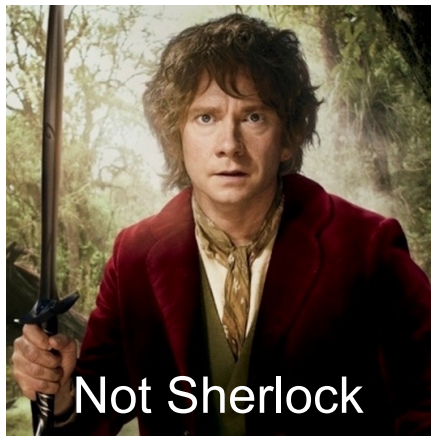
Up until the 20th century *Philosophy of Science* was mostly:

a) Part of wider philosophical investigations of knowledge (*epistemology*)
b) A hobby for retired scientists

Things change in 20th century as scientific successes boomed

Deductive Inference

Once upon a time, the best method to knowledge was ***Deduction*** *(Not Sherlock Holmes!)*



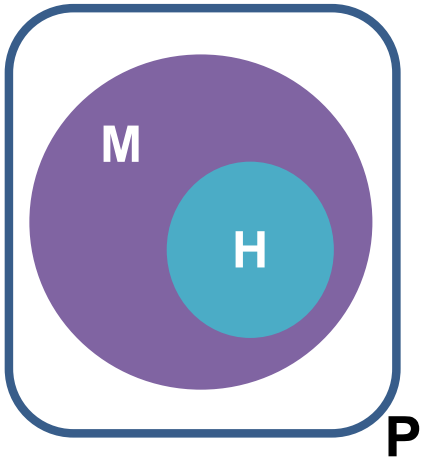
Steps of inference with certainty

- Each move as justified as the last (conserving)

→ Validity: impossible for premises true and conclusion false

For example:

P1: All humans are mammals	$P1\colon \forall x, Hx \rightarrow Mx, x \in P$
P2: Socrates is a human	$P2\colon Hs$
→ C: Socrates is a mammal	→ C: Ms



Rules like "*All Xs are…*" denote a list of *individual facts*

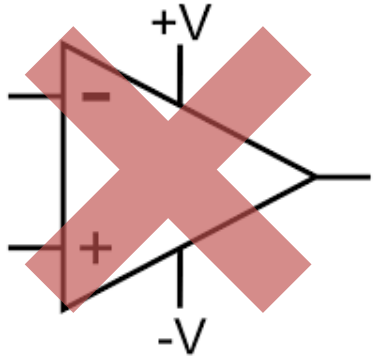
- $(Ha \wedge Ma),\ (Hb \wedge Mb), \ldots, (Hs \wedge Ms)$
- The conclusion is already in the premise (in the list)

There is no *new* information

- Though possibly you appreciate something you had not before

Classical Science

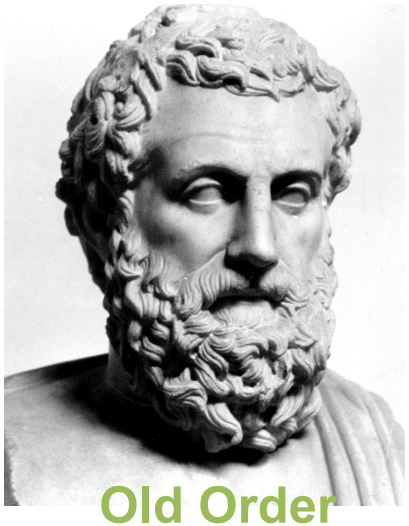
Deductive reasoning is strong but limited
- A *non-amplicative* tool of mathematical inference
- Limited use in empirical setting (and life!)



Where's the empirical bit?
- Collecting data

After a formula/rule is used to *order* the collection
- There is no rule for predictions beyond data

Aristotelian view of science
- Ordering of information according to some ideal scheme
  - Not necessarily empirical


Old Order

Doesn't meet the modern expectation of new knowledge

It's not that this isn't useful, but it isn't the whole story of modern science
- Of course, collecting data is a big part of science
- But scientists do more than just tell us about last week

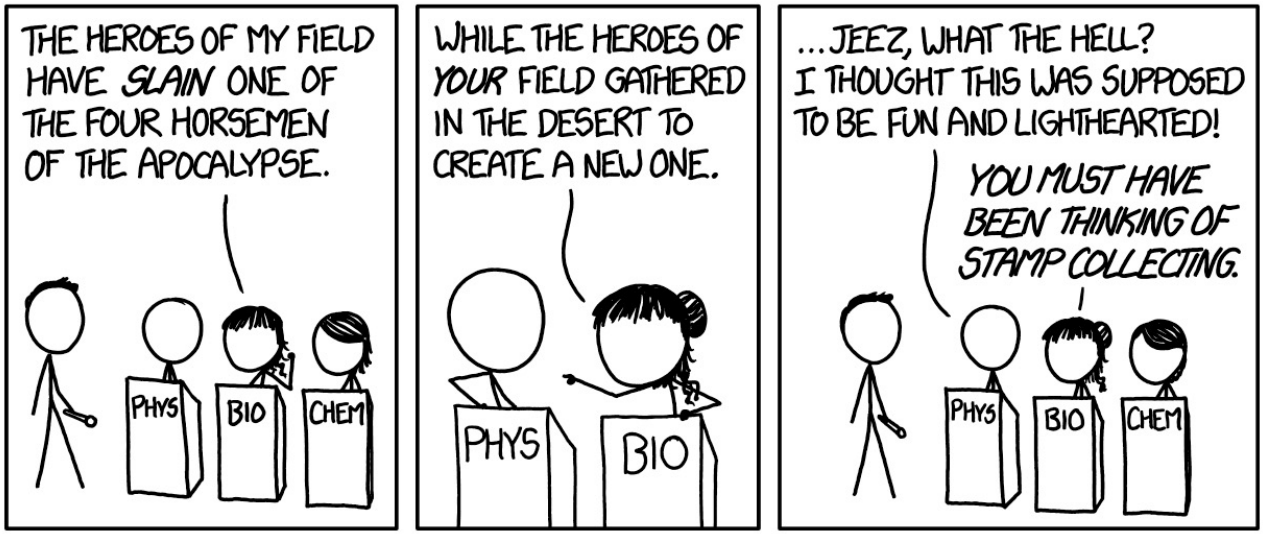
Take a moment…

Science clearly does have an ordering aspect to it

Some patterns seem obvious in data, others can take years to fathom

Ernst Rutherford, is attributed the quote: *"All science is either physics or stamp collecting"*

- There's a distinction here between putting things in order and doing something *physicsy (read: especially sciency)*
- This is an example of *Imperialism* in physics



Knowledge and Observation

Modern Philosophy of Science has two contrasting approaches of analysis

- Key difference lies in where *experience* plays a role

Induction – goes back to beginning of modern era (Bacon ca. C17th)
From observation *to* theory:

- Begin with data
- Identify regularities/patterns/correlations
- Characterise in law



Empirical input

Hypothetico-Deduction – really developed in twentieth century
To observation *from* theory:

- Generate hypothesis of regularities/patterns/correlations
- Predict novel phenomenon by deduction
- Test prediction



Empirical input

Each emphasises *empirical* input but in different parts of method

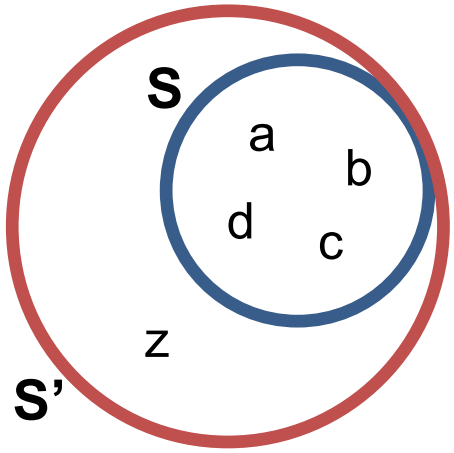
Inductive Inference

Deductive inference is robust but often uninteresting
Inferring new information is more exciting & useful→ *Amplicative inference*

Standard example:

P1: all observed frogs are green
P2: all frogs are green
P3: there is a (new) frog
C: the (new) frog is green

$P1: \forall x, Fx \rightarrow Gx, x \in S$
$P2: \forall x, Fx \rightarrow Gx, x \in S'$
$P3: Fz, z \in S'$
→ C: Gz



The step form P1 to P2 is an *inductive leap*

- The law goes **beyond the data collected**
- An inference is made to something yet to be observed

C is *deduction* from P2 and P3

Asymmetry to inductive arguments

- one counter-example defeats all corroborations
- Uncertainty is the price of ambition

(aside: who decides the *basic* observations anyway?)



Various Inferences

However, *Inductive reasoning* is **very common** and comes in various forms

- ***Enumerative* induction**: predicting further event(s)
  Suppose there are examples of some regularity, e.g. All ravens so far seen are black
  - *Weak*: the next raven seen will be black
  - *Strong*: all ravens are black



- ***Eliminative* induction**: finding a common cause
  Contrast where the phenomena was seen and not seen, e.g. I am grumpy when hungry, I'm happy when fed
  - Food keeps me happy

- ***Analogical Induction***: from (observed) properties to others (unobserved)
  Sets of properties are known to be shared so more are expected, e.g. Irn-Bru looks orange so it should taste citrusy



Each variety assumes a *Principle of Uniformity of Nature*

- Unobserved things will be like observed things

Riddles of Induction I

(Old) Riddle of Induction: Hume (1711-1776)

- Critique of *Principle of Uniformity of Nature* (**PUN**) as **not an empirical principle**

P1: Many observations: e.g. 99 ravens are black
P2: PUN
P3: There's another (new) raven
→C: It will be black

How do we know PUN is justified?
P1: Previous uses of PUN has worked well
P2: PUN
P3: New case of PUN
→ C: New PUN case will work well

Induction is used to justify *induction* (question begging)

- PUN worked yesterday → PUN will work tomorrow

Arguably: PUN is not an empirical principle



Difficult Scotsman



Unemployed Philosopher

Riddles of Induction II

New Riddle of Induction: Goodman (1906-1998)

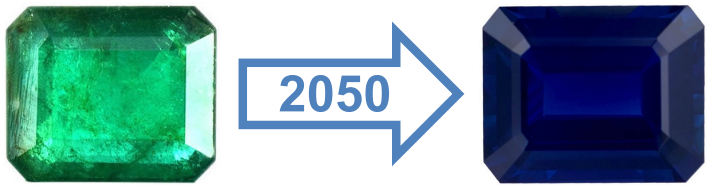
- Point out a propensity for particular kind of Scientific knowledge

Consider Scientific laws have the form: All Xs are G

- But not everything of this form is a law: All Beatles were Scouse

Suppose a property *Grue:* green until 2050 and blue after 2050

All emeralds are *green*
All emeralds are *grue*
} Agrees with experience (so far)



Is this a reasonable law? Is *Grue* a reasonable property?

- Intuition of laws referring to *"natural"* properties
- Grue does not look *"natural"* due to time dependence
- Goodman redefines Green: *grue* until 2050 and *bleen* after 2050
  - All properties can be defined in this *"unnatural"* way

Where does the intuition for *"natural"* properties come form?
(aside: who gets to decide what's natural?)

Arguably: Property & law definitions involves non-empirical principles

Induction Summary

Hume's and Goodman's riddles are used to show that **Induction *alone* is not *The* method of Science**

- Other principles are required to make the *right kinds* of induction
- The charge is that **these principles are not found empirically**

It's not that these are not useful, or used in science and in general

- Of course, we assume some uniformity in nature
- Of course, we make use of laws with reference to atemporal properties

But this isn't the whole story of the Scientific Method

- How do we know when to make assumptions?

Contra Newton: *Hypotheses non fingo*

- **Some hypotheses/principles are required to get Inducive inferences going**

Can another method do better?



Un-framed (apparently)

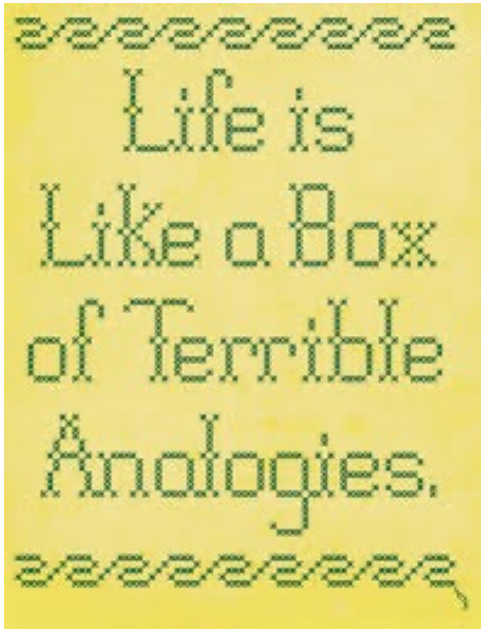
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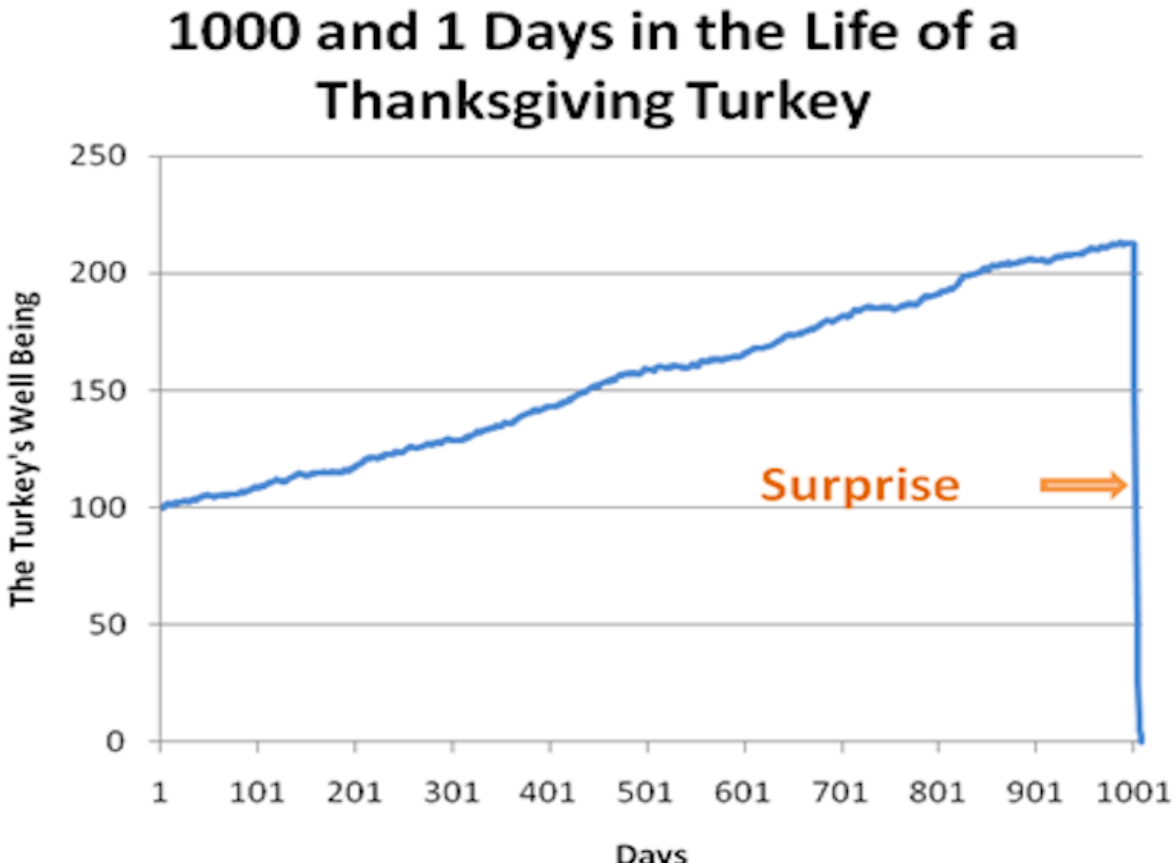
Take a moment…

Arguments from resemblance are *extremely* powerful

- *Induction* expands your understandable world
- *Elimination* helps fix praise/blame to phenomena
- *Analogies* make the unfamiliar more familiar

But beware…





Hypothetico-Deductive Model

Seems difficult to describe rational theory development (e.g. Induction)

- Let's park it! - question for another day / who cares?

New strategy: separate invention from justification

Theories can come from any inspiration, what counts is testing

- Empirical arbitration by Nature → *empirical* bit!



Hypothetico

- Come up with a hypothesis, any way you like

Deduction

- Deduce the consequences, especially novel phenomena

Consistency test (easy): Can it account for known phenomena?

Novel test: Can it predict a new phenomenon? (Verificationism)

Goal: **Construct a test of hypothesis based on novel prediction**

- Pass – confirmation/corroboration
- Fail - falsification

Confirmation Paradox

More evidence is more certain

- Hempel's (1905-1997) Paradox

Consider the testable hypothesis:
All ravens are black



$Rx \cap Bx$



$-Rx \cap -Bx$

Intuitively each observation raises belief/confidence

What about non-ravens?

- All raven-things are black-things ⇔ All non-black things are non-raven-things

  $(\forall x, Rx \rightarrow Bx) \leftrightarrow (\forall x, \neg Bx \rightarrow \neg Rx)$
- Therefore, *any* thing which is *not black and not a raven* is a confirmation, e.g. my shoe (or your shoe, if it's not black)

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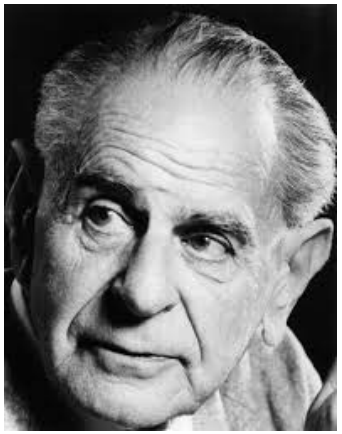
Falsification

Karl Popper (1902-1994)

Problems of confirmation are not easily solved

- Not easy to find threshold of data to justify confirmation

Popper's insight was to flip the problem:
Science is not a method of *confirmation*; science is a method of *falsification*



Truly Popper

There is no confirmation, only corroboration

- Agreement with prediction is <u>not</u> proof and has little (epistemic) value

Theories cannot be proved, but they can be refuted

- A failure in prediction defeats a theory
- Certainty in failure is better than no certainty at all



Scientific method remains a method of *progress*

Bonus: Popper returns a principle of *demarcation*

A theory without *falsification* (in principle) **is not scientific**

→ E.g.: astronomy in; astrology, Marxism, psycho-analysis out

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Falsificationism²

Popper knew & the history of science shows defeated theories can continue

- Replacements are required

E.g. Classical Physics refuted before advent of QM

- Utility may encourage persistence

E.g. Newtonian physics still goes far in much of terrestrial engineering; Relativity is not required



Relatively o.k.

Does this falsify Falsificationism?

- Popper gave more details and argued it would still do

What about cases where rival theories can *both* explain and predict equivalently (e.g. Copenhagen QM and Bohm's model)?

- Both had equivalent empirical accuracy
- Does the first theory always win by timing alone?

Are there other criteria, e.g. simplicity, utility, legacy to help choose

- If so, is empirical accuracy is just one of many desirable traits
- More tomorrow…

Recap: PoS

The story so far…
Looking for an *empirical method* in Science to justify high status

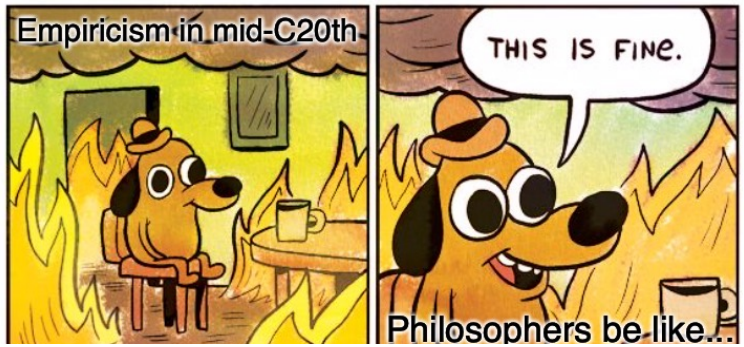
- *Ontological authority* from *epistemic privilege*

- Deduction does not do enough: need *amplicative* inference

- *Inductivism* seems relevant but incomplete
  - Need additional principles to read data and guide inferences
  - Some reason to think these are not *empirical*

- *Hypothetico-Deductive* method moves *empirical* aspect to testing
  - Invention of a theory is separate from justification
  - Confirmation has problems with how to deal with trivial agreement
- *Falsification* saw limitations of agreement
  - Use falsification as a guide for progress
  - Get *demarcation* principle back

Ok, some questions remain but at least there's still progress right … right?



Caveat and next time…

WARNING

- To tell this story I have omitted many details
- Almost everything mentioned here is truncated and much more can and has been said
- Other views are available



Next Time

Beyond Philosophy: Science Studies gets empirical!

Featuring:

- More History of Science
- Some Sociology of Science
- And the The Science Wars



Post-script

*Some** bits I missed out

*there's much more

Vienna Circle

Influential group of (German speaking) academics – scientists interested in philosophy and philosophers interested in science



Existed for ~10 years in interwar period with Modernist agenda to reform philosophy (of science) known as *Logical Positivism*

- Exclusive focus on *empirical* nature of science
- Reject previous thought – theology, meta-physics, non-empirical theories
- Recast philosophy as *meta*-subject – role to decide what kind of statements are scientific/non-scientific (sensible/non-sensible), rather than contribute new knowledge

Before WWII most of the (Jewish and Leftist) group fled to UK & USA

- This had a huge impact on Anglophone philosophical tradition
- Contributing to Analytical/Continental branching in philosophy

Influence on next generation of philosophers of science, inc. Quine and Kuhn